

LETHBRIDGE CDA, AB, Canada

WMO: 712430

Lat: 49.700N

Lon: 112.770W

Elev: 2986

StdP: 13.18

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.8	-11.9	-24.7	1.6	-17.6	-18.2	2.3	-11.7	35.7	40.5	32.1	42.2	5.5	200	0.676

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.5	89.2	62.6	85.6	61.6	82.1	60.7	66.3	81.3	64.4	79.6	62.5	77.6	10.2	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.2	90.2	72.5	58.9	82.9	70.4	56.7	76.5	67.9	32.7	81.2	31.1	79.8	29.6	77.5	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
29.7	26.4	23.4	-25.9	94.5	6.6	3.0	-30.6	96.6	-34.5	98.3	-38.2	100.0	-43.1	102.1
			-26.2	70.6	6.6	2.6	-30.9	72.5	-34.7	74.0	-38.4	75.4	-43.2	77.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.6	22.1	24.5	31.9	42.7	52.6	59.9	66.5	65.1	56.5	44.6	32.3	23.4
	DBStd	19.51	17.28	15.39	14.39	9.16	8.18	6.03	5.48	6.08	8.41	9.65	13.63	15.25
	HDD50	4069	866	714	563	247	67	4	0	1	35	214	535	823
	HDD65	8020	1330	1134	1025	670	392	173	47	75	274	632	980	1288
	CDD50	1741	1	1	4	27	147	301	511	468	228	48	5	0
	CDD65	217	0	0	0	0	6	21	93	78	18	1	0	0
	CDH74	3523	0	0	1	14	176	324	1326	1253	417	12	0	0
	CDH80	1192	0	0	0	1	38	75	473	470	134	1	0	0
	WSAvg	10.0	11.1	10.0	10.5	10.7	10.1	9.7	8.5	7.9	8.8	10.4	11.0	11.1
	PrecAvg	16.7	0.9	0.6	1.1	1.6	2.1	2.6	1.8	1.7	1.8	0.7	0.7	0.9
Precipitation	PrecMax	30.2	2.3	1.5	3.1	6.7	5.3	6.7	4.8	5.6	5.5	1.8	2.0	2.3
	PrecMin	9.6	0.2	0.1	0.2	0.0	0.4	0.1	0.3	0.2	0.2	0.0	0.0	0.2
	PrecStd	4.4	0.5	0.3	0.6	1.3	1.2	1.7	1.3	1.3	1.4	0.5	0.6	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.1	57.5	64.9	75.9	84.4	87.8	94.0	93.4	89.7	75.5	64.3	54.6
		MCWB	43.3	42.3	46.7	52.5	58.1	63.3	64.5	62.3	59.5	53.4	48.1	41.8
	2%	DB	50.6	52.2	59.5	69.3	79.1	81.5	89.1	89.3	83.9	69.5	58.1	49.7
		MCWB	40.4	40.4	44.8	49.5	56.4	60.7	64.6	61.6	58.1	51.7	45.2	39.5
	5%	DB	47.0	48.0	55.3	63.7	73.9	77.6	85.4	85.3	78.9	64.3	53.7	46.2
		MCWB	38.3	37.8	42.4	46.8	53.9	59.4	63.6	60.6	57.0	49.6	42.8	37.3
	10%	DB	43.8	44.2	50.8	58.9	69.1	73.7	81.5	81.5	73.3	59.7	49.6	43.0
		MCWB	36.3	35.4	40.0	44.5	51.8	57.3	62.6	60.0	54.8	46.8	40.1	35.4
	0.4%	WB	44.4	43.9	48.4	53.5	59.9	66.7	70.8	66.4	62.3	55.6	50.2	43.4
		MCDB	54.3	55.2	62.5	73.1	79.2	81.0	85.2	84.1	82.4	71.8	62.2	52.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	40.9	40.9	45.5	50.3	57.2	63.2	67.6	64.3	59.6	52.6	46.0	40.2
		MCDB	49.5	50.8	57.7	66.6	76.1	77.2	82.1	81.6	79.4	67.3	56.7	48.7
	5%	WB	38.7	38.3	42.9	47.7	55.1	60.8	65.6	62.6	57.6	50.0	43.2	37.7
		MCDB	46.4	47.4	54.0	62.3	71.4	73.9	80.1	79.1	75.9	63.3	52.8	45.4
	10%	WB	36.5	35.8	40.3	45.1	52.9	58.6	63.9	61.1	55.6	47.4	40.6	35.4
		MCDB	43.7	43.8	50.2	57.7	66.9	70.9	78.4	78.0	71.7	58.8	49.3	42.9
Mean Daily Temperature Range	5% DB	MDBR	19.3	20.2	20.7	21.9	22.7	21.3	25.5	26.9	24.6	21.5	18.9	18.5
		MCDBR	19.8	23.3	26.9	31.3	31.5	28.6	32.6	35.3	35.4	28.7	23.2	19.5
		MCWBR	13.1	14.4	15.3	15.9	14.0	12.6	13.8	13.6	14.6	14.7	13.8	12.6
	5% WB	MCDBR	20.0	23.6	26.5	29.3	29.1	25.4	28.3	30.1	32.4	27.7	23.0	19.8
		MCWBR	13.7	15.2	15.7	15.3	13.3	12.5	13.8	12.7	13.9	14.5	14.3	13.4
Clear-Sky Solar Irradiance	taub		0.250	0.259	0.283	0.319	0.332	0.330	0.333	0.350	0.312	0.280	0.256	0.234
	taud		2.458	2.491	2.484	2.392	2.407	2.461	2.468	2.396	2.474	2.542	2.505	2.458
	Ebn at Noon		264	289	298	294	294	294	292	281	283	277	259	255
	Edh at Noon		18	23	28	34	35	34	33	34	28	22	17	16
All-Sky Solar Radiation	RadAvg		393	710	1111	1536	1830	2035	2236	1778	1262	786	438	310
	RadStd		32	56	92	103	152	147	115	109	106	80	35	30

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (67 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page